

PROJECT DOCUMENTATION

ON

**AUTO TICKET CLASSIFICATION USING
FLOW DESIGNER IN SERVICENOW**

Submitted in partial fulfillment of the requirements for the Naan Mudhalvan Skill Development Programme



NAAN MUDHALVAN
Skill Development Programme

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1. Introduction

Project Title

Auto Ticket Classification using Flow Designer in ServiceNow

Team Members

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2. Project Overview

Purpose

The purpose of this project is to automate the classification and assignment of service tickets using Flow Designer in ServiceNow. In traditional IT service management systems, tickets are often classified and assigned manually, which can lead to delays, incorrect categorization, and increased workload for support teams.

This project addresses these challenges by implementing an automated workflow that identifies ticket categories based on predefined conditions and assigns tickets to the appropriate support groups.

The solution aims to reduce manual effort, improve ticket assignment accuracy, and enhance the overall efficiency of IT support operations.

Features

- **Automatic Ticket Classification:** Classifies tickets based on predefined conditions and ticket information.
- **Automated Assignment:** Routes tickets to the appropriate support groups.
- **Flow Designer Integration:** Uses triggers, conditions, and actions to automate ticket processing.
- **Real-Time Processing:** Initiates the configured workflow when a new ticket is created.
- **Reduced Manual Effort:** Minimizes the need for manual ticket categorization and assignment.
- **Consistent Classification:** Applies the same classification rules across different tickets.
- **Error Handling:** Supports default handling for tickets that do not match predefined conditions.
- **Testing and Validation:** Enables verification of classification and assignment through Flow Designer execution results.

3. Architecture

System Architecture

The project is developed entirely within the ServiceNow platform. It uses Flow Designer to automate the processing of service tickets without requiring a separate frontend or backend application.

The system consists of the following components:

- **Employee / End User:** Submits a service ticket with issue details.
- **ServiceNow Incident Management:** Receives and stores the ticket information in the Incident table.
- **Flow Designer:** Executes the configured trigger, conditions, and actions to classify the ticket.
- **Automatic Classification and Assignment:** Determines the ticket category and assigns the appropriate support group.
- **IT Support Team:** Receives the classified ticket for further processing and resolution.

Platform

ServiceNow: Provides the environment for ticket creation, data storage, workflow configuration, and service management.

Flow Designer

Flow Designer is the primary automation component of the project. It is configured using:

- Triggers to initiate the flow when a ticket is created.
- Conditions to identify the appropriate ticket category.
- Actions to update ticket fields and assignment groups.
- Flow execution details to monitor the results.

Database

The project uses the ServiceNow platform's built-in database.

Table / Component	Purpose
Incident (incident)	Stores ticket details, including short description, category, priority, and assignment group.
User (sys_user)	Maintains user information associated with service tickets.
Group (sys_user_group)	Stores support group information used for ticket assignment.
Flow Execution Data	Provides execution details for monitoring and validating automated workflows.

The project uses existing ServiceNow tables and does not require a separate MongoDB database.

4. Setup Instructions

Prerequisites

The following are required to implement and test the project:

- A ServiceNow Developer account.
- Access to a ServiceNow Developer Instance.
- A modern web browser such as Google Chrome or Microsoft Edge.
- Basic knowledge of ServiceNow Incident Management.
- Familiarity with Flow Designer.
- Appropriate permissions to create and configure flows.
- Internet connectivity.

Installation and Configuration

Step 1: Access ServiceNow Developer Instance

Visit the ServiceNow Developer Portal and sign in or create an account to access a developer instance.

Step 2: Open the Instance

Launch your assigned ServiceNow instance and log in using your authorized credentials.

Step 3: Navigate to Flow Designer

1. Open the Application Navigator.
2. Search for Flow Designer.
3. Open the Flow Designer module.
4. Select the option to create a new flow.

Step 4: Configure the Flow

1. Provide a suitable name for the flow.
2. Select the appropriate trigger for ticket creation.
3. Configure conditions based on ticket information.
4. Add actions to update ticket fields.
5. Configure the assignment group based on the classification conditions.

6. Save and activate the flow.

Step 5: Test the Flow

1. Create sample service tickets.
2. Verify the ticket classification.
3. Check the assigned support group.
4. Review the flow execution details.
5. Confirm that the workflow completes successfully.

5. Project Structure

Unlike a traditional software application, this project does not contain separate React client and Node.js server folders. The implementation is organized within the ServiceNow platform.

Logical Project Structure

```
Auto Ticket Classification System - ServiceNow/
├── 1. Ideation Phase/
│   ├── Brainstorm_Idea_Prioritization.pdf
│   ├── Define_The_Problem_Statement.pdf
│   └── Empathy_Map_Canvas.pdf
├── 2. Requirement Analysis/
│   ├── Customer Journey Map.pdf
│   ├── Data Flow Diagrams and User Stories.pdf
│   ├── Solution Requirements.pdf
│   └── Technology Stack.pdf
├── 3. Project Design Phase/
│   ├── Problem Solution Fit/
│   │   └── Problem - Solution Fit.pdf
│   ├── Proposed Solution/
│   │   └── Proposed Solution.pdf
│   └── Solution Architecture/
│       └── Solution Architecture.pdf
├── 4. Project Planning Phase/
│   ├── Project Logic.pdf
│   └── Project Planning.pdf
├── 5. Project Development Phase/
│   ├── Performance Testing/
│   │   ├── Artificial Intelligence.pdf
│   │   ├── GenAI Functional Performance Testing.pdf
│   │   ├── GenAI_Functional_Performance_Testing.pdf
│   │   ├── Machine Learning.pdf
│   │   ├── Power BI.pdf
│   │   ├── Tableau.pdf
│   │   └── User Acceptance Testing.pdf
│   └── User Acceptance Testing/
│       └── UAT Report.pdf
├── 6. Project Documentation/
│   ├── Project Documentation.pdf
│   └── Project Report.pdf
└── README.md
```

This is a suggested GitHub documentation structure. The actual flow configuration and records are maintained within your ServiceNow instance.

6. Running the Application

Since this project is implemented on the ServiceNow platform, there are no local frontend or backend servers to start.

Steps to Run the Project

1. Log in to your ServiceNow Developer Instance.
2. Navigate to Flow Designer.
3. Locate the configured Auto Ticket Classification flow.
4. Ensure the flow is activated.
5. Navigate to the Incident or ticket creation form.
6. Create a new ticket with the required information.
7. Submit the ticket.
8. Allow the flow to execute.
9. Verify the automatically assigned category and support group.
10. Review the flow execution details to confirm successful processing.

Expected Output

The ticket should be automatically classified and assigned to the appropriate support group based on the configured conditions.

7. API Documentation

API Integration

The current project uses ServiceNow's built-in platform functionality and Flow Designer actions. It does not require a separately developed REST API or external backend service.

The system processes ticket information using configured triggers, conditions, and actions within ServiceNow.

Core Operations

Operation	Description	Method
Create Ticket	Creates a new service ticket in ServiceNow.	ServiceNow Form
Retrieve Ticket	Displays existing ticket information.	ServiceNow Form
Classify Ticket	Identifies the ticket category using configured conditions.	Flow Designer
Assign Ticket	Updates the assignment group automatically.	Flow Designer Action
Monitor Execution	Displays the flow execution status and details.	Flow Designer

Optional REST API Integration

ServiceNow provides REST APIs that can be used for future integration with external applications. However, external API integration is not required for the current implementation.

8. Authentication and Authorization

The project uses ServiceNow's built-in authentication and access control mechanisms.

Authentication

- Users access the ServiceNow instance using authorized login credentials.
- Authentication is managed by the ServiceNow platform.
- Only authenticated users can access protected application functionality.

Authorization

Access to project functionality depends on assigned user roles and permissions.

- **Administrator:** Can configure the instance, manage users, and modify workflows.
- **Developer:** Can create and maintain Flow Designer workflows and configure project components, subject to assigned permissions.
- **IT Support User:** Can access and process tickets according to assigned permissions.
- **End User:** Can create and view tickets based on available access rights.

Security

The project relies on ServiceNow's access control rules (ACLs) to protect records and restrict unauthorized access.

9. User Interface

The project uses the standard ServiceNow user interface for ticket creation, flow configuration, and execution monitoring.

Main Interface Components

1. **ServiceNow Dashboard:** Provides access to available modules and applications.
2. **Incident Form:** Allows users to create and submit service tickets.
3. **Flow Designer Interface:** Provides a visual environment for configuring automated workflows.
4. **Condition Builder:** Defines the rules used for ticket classification.
5. **Flow Execution Details:** Displays the execution status and results.
6. **Ticket Record:** Shows the final ticket information, including its category and assignment group.

Screenshots

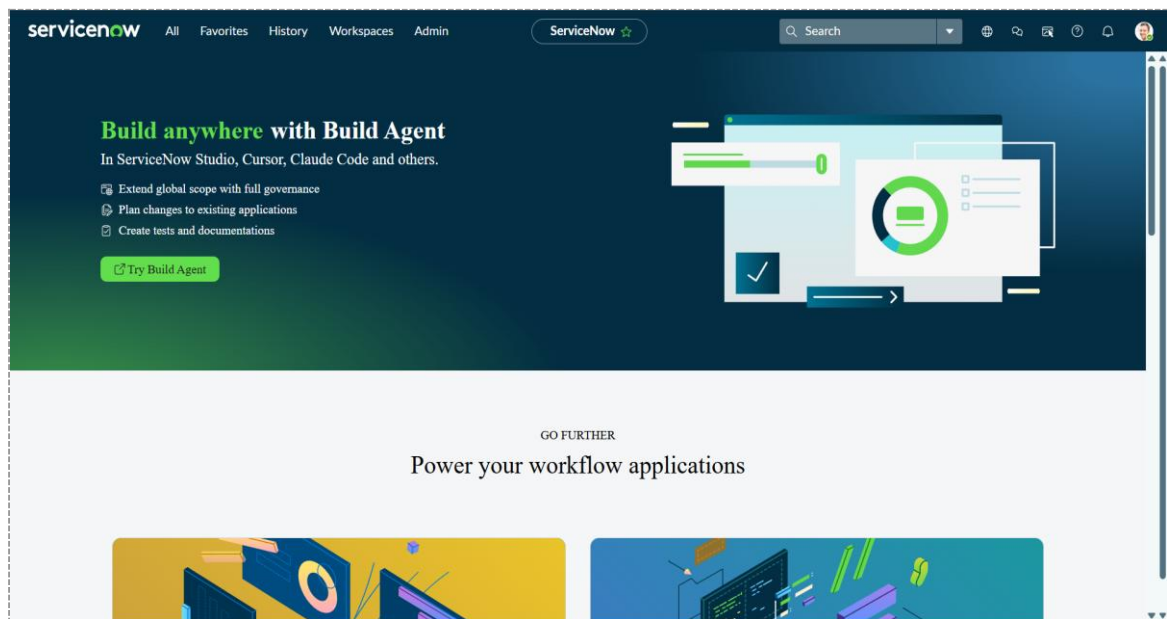


Figure 1: ServiceNow Developer Instance

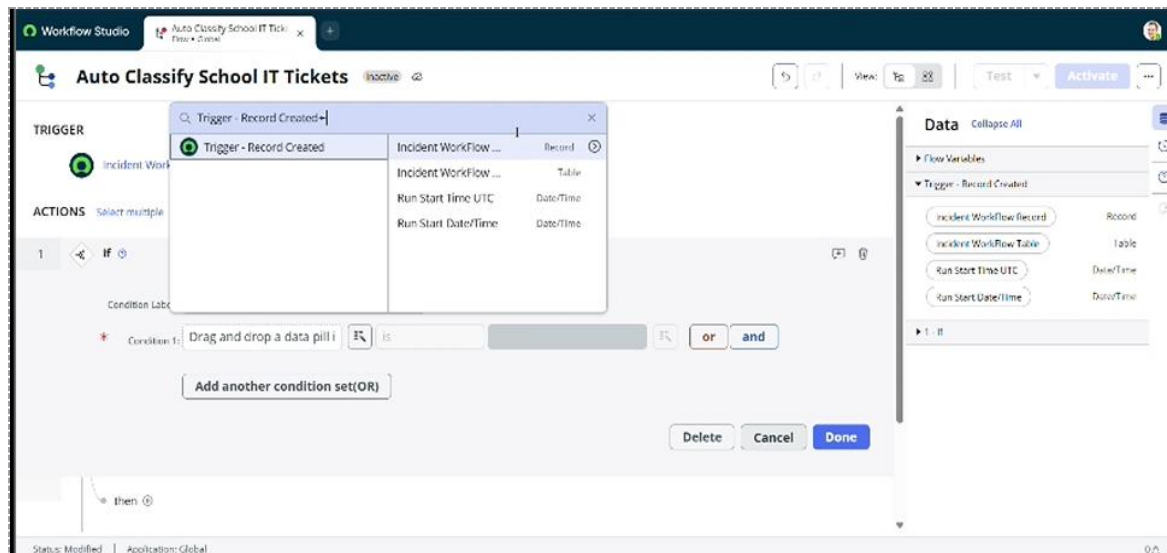


Figure 2: Ticket Creation Form

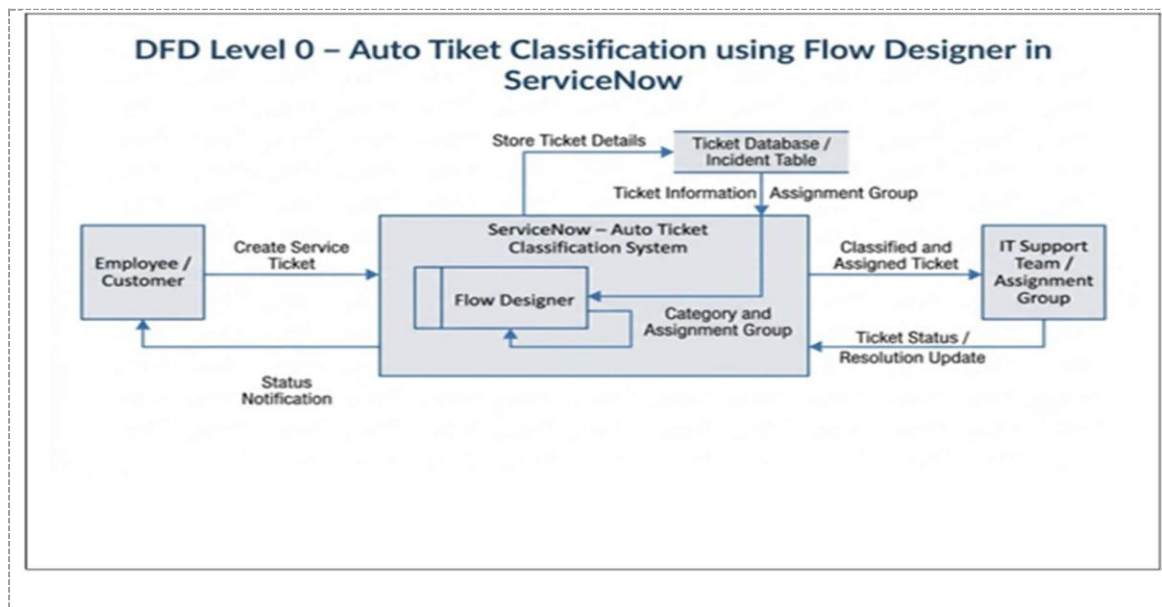


Figure 3: Flow Designer Workflow

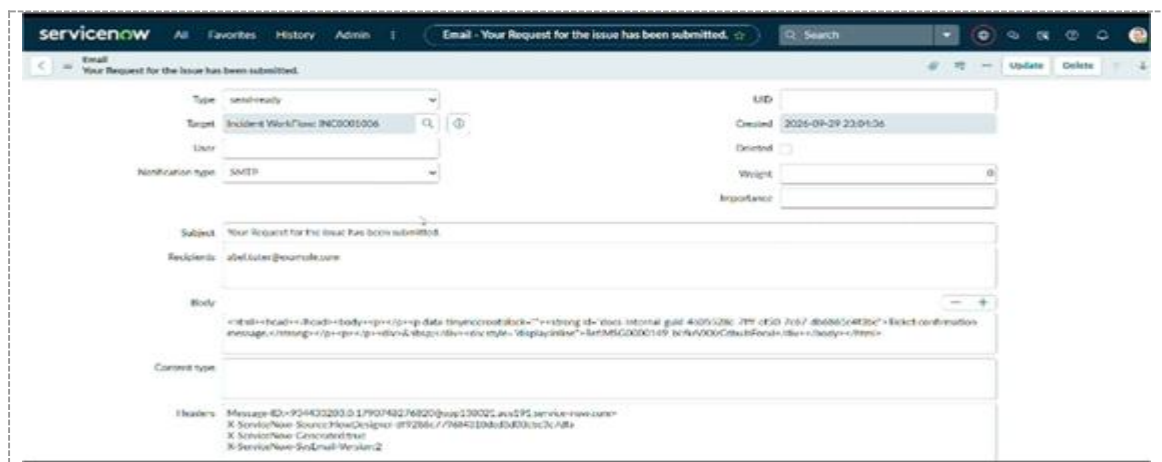


Figure 5: Final Ticket Output

10. Testing

Testing was performed to verify the functionality, reliability, and performance of the automated ticket classification system.

Testing Types

Functional Testing

- Ticket creation validation.
- Automatic ticket classification.
- Assignment group selection.
- Flow trigger validation.
- Condition verification.
- Unclassified ticket handling.

Performance Testing

- Flow execution time.
- Multiple ticket processing.
- Flow reliability.

User Acceptance Testing (UAT)

The system was tested to ensure that it meets the requirements of employees and IT support teams.

Test Results

Testing Type	Total Test Cases	Passed	Failed
Functional Testing	5	5	0
Performance Testing	3	3	0
Total	8	8	0

100% Test Case Pass Rate

Testing Conclusion: All 8 reported test cases were successfully completed. The automated classification and assignment workflow functioned as expected during testing.

Demonstration Steps

1. Open the ServiceNow Developer Instance.
2. Navigate to the ticket creation form.
3. Enter the ticket details.
4. Submit the ticket.
5. Observe the automatic execution of Flow Designer.
6. Verify the assigned ticket category.
7. Check the assignment group.
8. Review the final ticket record.

Demo Video

<https://drive.google.com/drive/folders/1k74T4fTFXg1D21zHFwUSWsYuT8qtD4nV?usp=sharing>

GitHub Repository

<https://github.com/uchihatanziro-cell/Auto-Ticket-Classification-using-Flow-Designer---ServiceNow.git>

11. Known Issues

During the reported testing, no major functional issues were identified.

The following considerations apply to the current implementation:

- Ticket classification depends on the conditions configured in Flow Designer.
- Tickets with unclear or incomplete descriptions may require manual review.
- New ticket categories may require additional classification conditions.
- Changes to assignment groups may require corresponding updates to the workflow.
- Flow execution depends on the availability and configuration of the ServiceNow instance.

These are considerations for maintaining and extending the system rather than confirmed defects.

12. Future Enhancements

The following improvements can be considered for future versions of the project:

1. **AI-Based Classification:** Integrate machine learning or ServiceNow AI capabilities to classify tickets using natural language understanding.
2. **Improved Classification Rules:** Expand the existing conditions to support a wider range of ticket descriptions and categories.
3. **Priority-Based Assignment:** Automatically determine ticket priority based on urgency and impact.
4. **Automated Notifications:** Notify users and support teams when tickets are created, classified, assigned, or updated.
5. **SLA Monitoring:** Integrate service-level agreement monitoring to help track ticket response and resolution times.
6. **Advanced Reporting:** Develop reports to monitor ticket volume, classification accuracy, assignment patterns, and resolution time.
7. **Integration with Other Systems:** Connect ServiceNow with external applications through REST APIs where required.
8. **Continuous Improvement:** Review ticket classification results and refine conditions based on user feedback and operational needs.

Conclusion

The Auto Ticket Classification using Flow Designer in ServiceNow project demonstrates how workflow automation can improve IT service management by reducing manual ticket classification and assignment.

By using Flow Designer triggers, conditions, and actions, the system automatically identifies ticket categories and routes service requests to the appropriate support groups.

The project was tested using functional, performance, and user acceptance testing scenarios, with all reported test cases passing successfully.

The implementation provides a foundation for improving ticket management efficiency and can be extended in the future with AI-based classification, advanced reporting, and additional automation capabilities.